

Green AI: Assessing the Carbon Footprint of Fine-Tuning Pre-Trained Deep Learning Models in Medical Imaging

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Abstract—Artificial Intelligence (AI) is at the forefront of today's research trends, particularly in deep learning. The prevailing trend in designing AI systems is based on the principle “the bigger, the better,” which focuses on achieving higher scores on benchmarks. However, this approach comes at a significant environmental cost. At the same time, reducing carbon footprint emissions is more crucial than ever. This study evaluates the environmental impact of fine-tuning the Google ViT model for medical image analysis. It also examines the impact of selecting the appropriate pre-trained model size, the influence of hardware architecture used for fine-tuning, and whether the choice of online providers affects the total emissions and energy consumption of the process. Using CodeCarbon, we calculated that each hyper-parameter fine-tuning experiment required about 0.18 kWh of energy to complete and produced 0.066 kg of equivalent CO₂ emissions. We also found that using different sizes of the pre-trained ViT model results in varying environmental impact and efficiency. Finally, we tested hardware over-sizing and discovered that it can increase the emissions produced

Index Terms—Green AI, deep learning, ViT model, carbon footprint, energy consumption, fine-tuning, medical image analysis

I. INTRODUCTION

Artificial Intelligence (AI) is acknowledged as one of the most important tools in healthcare, from predictions to new cures and innovative drugs. In medical image analysis, AI aids in accurate diagnosis, leading to more efficient medical practices and better patient outcomes. The core of these systems is based on deep learning models, many of which are built upon the transformer model [1], such as Google's Visual Transformer model (ViT) [2], or convolutional neural networks (CNN) like the You Only Look Once (YOLO) model [3].

Both of these model types require vast datasets for training (the amount of digital data is expected to reach 175 ZB by 2025 [4]) and also enormous hardware resources with an intense carbon footprint [5] [6]. Furthermore, the main principle of these models is based on the simple rule “the bigger, the better,” which has led the majority of the research community to focus more on accuracy rather than efficiency, pursuing higher scores on benchmarks and crafting more

complex systems with larger sizes to achieve these scores. Consequently, their environmental footprint increases exponentially [7]. This increased energy demand due to AI is evidenced by the 2024 environmental report of Google [8], which shows a 13% increase in t CO₂ emissions compared to 2022, and also by the Microsoft 2024 sustainability report [9], which reports a 29.1% increase in emissions from the 2020 baseline. Even though both companies are targeting a future zero energy footprint, in practice, this means they purchase emissions from power production that uses renewable energy, which differs from reducing the actual production in their systems. The real goal should be the actual reduction of AI's environmental footprint.

These challenges of “Red Computing” (trying push the boundaries of AI while also threatening the environment and natural resources) have led to movements that study how AI can be more environmentally friendly. These movements are named as “Green AI”, a term that first appeared in the research [11] that proposes the “Green Computing” framework as a solution to this problem. According to the framework, four pillars must be followed from the design through the production stages of AI models:

- 1) Measure the Greenness
- 2) Energy-Efficient AI,
- 3) AI for Sustainability, and
- 4) Energy-Efficient Computing Systems

Our case study is focused on a visual transformer model and belongs to the first of the above category of the framework that suggests to measure:

- 1) Running time,
- 2) Power consumption
- 3) Model size/FLOPS, and
- 4) Carbon emissions

A number of available tool-kits allows these measurements, such as CodeCarbon [12], Carbontracker [13], and Green Algorithms [14], along with comparison studies [15] [16].

This study was conducted as a part of learning how to fine tune vision transformers for medical usages. Since vision

application and models increase in every day applications, measuring the environmental impact of the process was set as a primary scope trying to understand the part of the process that have impact on the energy and the produced emissions. The scientific contribution of this case study focuses on answering the following questions regarding the fine-tuning process of a ViT model for medical image analysis, which has not been addressed up to the time of this study:

- 1) What is the environmental impact of fine-tuning the specific use case in a ViT model?
- 2) What is the environmental impact of selecting a different pre-trained ViT model for this use case?
- 3) What is the impact of system over-sizing on the environmental footprint in this specific use case?
- 4) How does the selection of an online server provider affect the emissions and the consumed energy?

In the literature, we found one publication related to the environmental impact on the training process of a Google Visual Transformer [10]. In that study the energy consumption demands for training five different sizes of the model were evaluated on an AWS server for the ImageNet [17] dataset. Our case study is different due to studying the fine-tuning process of the ViT model, not the training process. Also, in our case with a much smaller dataset for medical image analysis.

This case study follows the following structure: Chapter II presents the related work on “Green AI” for machine learning. Chapter III presents the methodology this case study follows so other researchers can reproduce the experiment. Chapter IV presents and analyzes the results. Chapter V provides the conclusions of this case study. Chapter VI discusses the findings. Chapter VII suggests future work.

II. RELATED WORK

In this part we organised the related research we found, mainly in the Scopus database, presenting the active fields of research.

Surveys: Numerous surveys have been conducted covering the state of the art, research trends, advancements, tools, and future research opportunities [6] [7] [18] [19] [20]. In these surveys many frameworks providing guides, best practices, and metrics presented on how to measure AI’s environmental footprint. Most of these frameworks highlight the need to include the environmental footprint in benchmark scores, efficiency metrics and cost [22]. Some go further to include life cycle analysis calculations and models [23]. [24].

Green AI knowledge organization and categorization: In the review [25] research work has been conducted to recognize all the Green AI initiatives, that are organized into six major categories: cloud optimization, model efficiency, carbon footprinting, sustainability-focused AI development, open-source initiatives, and research and community. For each of the above the authors analyze the strengths, weaknesses, potentials and tools. In another review [26] green AI knowledge was categorized in an ontology model. This model focuses on energy consumption of published AI models. It is designed to model the metrics and tools, hardware and software settings,

cloud services and locations. Finally, we found two articles that analyse tools and metrics available for calculating environmental impact providing detailed insights [15] [16].

Frameworks: Two studies compare TensorFlow with the PyTorch frameworks applied to NLP and image task models. These studies report that TensorFlow performs better for training recommendation systems and computer vision models, while PyTorch excels in NLP categories [27]. Additionally, TensorFlow is found to be more energy-efficient during the training stage but less efficient in the inference phase. Negar Alizadeh et al. [28] report that their experiments did not identify a clear winner, with frameworks performing differently depending on varying models, batch sizes, and runtime environments.

Model Training and optimization: Two studies, focused on green AI and energy efficiency in NLP models, provide metrics and comparisons [29] but also examine how newer model technologies can reduce consumption. [33]. Related to the fine-tuning process, that is widely used in large language and vision models, the authors of [30] present a technique known as “Green-Trainer”. This technique is evaluated on three large language models (OPT, BLOOMZ, and FLAN-T5), and shows a 64% Green AI improvement in training without noticeable accuracy loss. It also highlights the potential for reducing computational costs while maintaining model performance. A similar research [32], focused on energy efficiency for automatic hyper-parameter optimization with Auto ML, found that a reduction of 28% in CO2 emissions was achieved by optimizing the hyper-parameters while also having as metric the minimization of energy consumption. Lastly, in a third publication [32] related to Green AI and Variational Autoencoders (VA), the researchers tried to estimate the performance of many VA models in terms of Green AI through the FLOPs metric (a hardware indented metric). However, they report that they did not manage to find a relation between the FLOPs and the real execution time.

Vision Applications: Experiments with Vision Transformers on the ViT model have been conducted on AWS servers for five different model sizes, showing that larger models have a larger impact [10]. Additionally, a lightweight version of I-ViT-T on Edge demonstrated a 2.9X energy savings with a reduction in performance by about 6% [44]. No other related works have been found on the fine-tuning process of ViT for medical image analysis. Therefore, our case study aims to provide experimental data to the research community.

III. METHODOLOGY

A. Research Design

This case study aims to fine-tune a Google Visual Transformer model for the task of medical image analysis, using a small dataset, and to estimate the environmental impact of this process. We adopted an experimental research design where we executed four experiments, each one targeting one of the specific research questions described in the introduction section.

TABLE I
GOOGLE'S ViT, PRE-TRAINED MODELS

Google Vision Model- model sizes						
	Model	Layers	Hidden Size	Attention Heads	Parameters	Classes
ViT-Base	google/vit-base-patch16-224	12	768	12	86 million	1000
	google/vit-base-patch16-224-in21k	12	768	12	86 million	21000
	google/vit-base-patch16-384	12	768	12	87 million	1000
	google/vit-base-patch32-384	12	768	12	88 million	1000
ViT-Large	google/vit-large-patch16-224	24	1024	16	307 million	1000
	google/vit-large-patch16-224-in21k	24	1024	16	307 million	21000
	google/vit-large-patch16-384	24	1024	16	308 million	1000
	google/vit-large-patch32-384	24	1024	16	309 million	1000
ViT-Huge	google/vit-huge-patch14-224-in21k	32	10280	16	632 million	21000

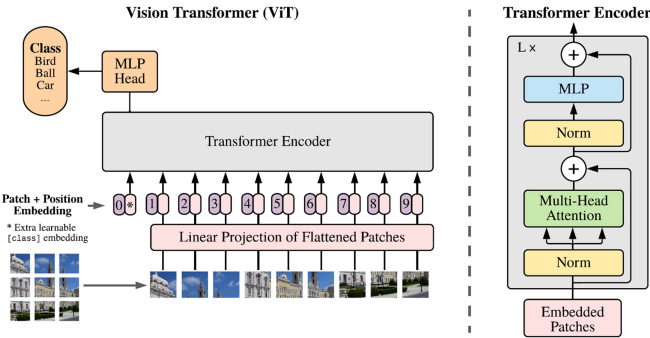


Fig. 1. Google's ViT model Architecture [2]

B. Dataset

For the fine-tuning process, we used a dataset of breast cancer ultrasound images (emre570/breastcancer-ultrasound-images) from the Hugging Face library [27]. This small dataset consists of a total of 780 images with three classes: benign, malignant, and normal.

C. Model selection

The selected model for this task is Google's Visual Transformer from the Transformers library of Hugging Face [35]. We used many versions of this pre-trained model, as we will present in the study "Table.I". The Python code created for the fine-tuning of the ViT model followed the Hugging Face tutorial on fine-tuning the ViT model [36].

The Visual Transformer-based model is inspired by BERT [37] and adapted to work with images instead of natural language processing. The main mechanism involves accepting an image as input, splitting it into a number of fixed-size patches, and then flattening each patch into a one-dimensional vector. These vectors are then converted into embedding vectors through a linear projection. Two additional embeddings are added: 1) The "Class token," which helps aggregate information across all patches for the classification process, and 2) the "Position embeddings," which assist with the proper positioning of the patches. The transformer model receives these vectors, where the attention mechanism calculates similarity scores between queries and keys to determine the relevance of each patch to the others. Finally, a normalization layer is applied, and a classification head is attached to perform the classification task, "Fig.1".

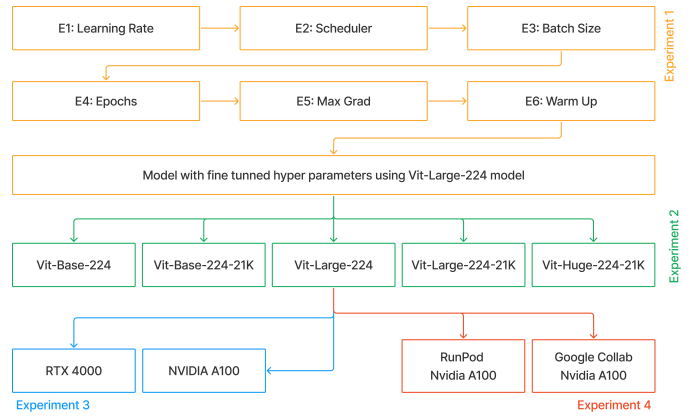


Fig. 2. Experiments Graph: E1 (Orange) hyper-parameter optimization. E2 (Green) comparison of ViT models. E3 (Blue) over-sizing impact. E4 (Red) server selection impact

TABLE II
EXPERIMENT 1: HYPER-PARAMETER OPTIMIZATION

Hyperparameter optimization energy consumption (without RAM impact)					
	Energy (kWh)	duration (hr)	Power (kW)	Emissions (kg CO2)	Emission Rate (kg CO2/hr)
E1: 1e-1	0.166	0.747	0.222	0.061	0.082
E1:1e-3	0.167	0.746	0.224	0.061	0.082
E1:1e-5	0.167	0.743	0.225	0.061	0.083
E1:1e-6	0.168	0.750	0.224	0.062	0.082
E2:Constant	0.168	0.748	0.224	0.062	0.082
E2:Cosine	0.168	0.752	0.223	0.062	0.082
E3: b16	0.164	0.728	0.225	0.060	0.083
E3: b32	0.164	0.686	0.239	0.060	0.088
E4:60 epoc	0.257	0.980	0.263	0.095	0.097
E4:100epoc	0.257	0.980	0.273	0.095	0.097
E5:MaxGrad 0.95	0.165	0.741	0.223	0.061	0.082
E5:MaxGrad 0.98	0.166	0.741	0.223	0.061	0.082
E6: WarmUp 100	0.166	0.748	0.222	0.061	0.082
E6: WarmUp 500	0.165	0.748	0.223	0.061	0.081
Total	2.507	10.836	3.232	0.923	1.185
Average	0.179	0.774	0.231	0.066	0.085

D. Fine-Tuning Procedure

To answer each research question set in the introduction, separate experiments were designed and executed "Fig.2".

In the first experiment (Experiment 1), the focus was set on finding the best hyper-parameter settings for this case study. Fourteen scenarios were run to fine-tune six hyper-parameters of the model, while the model type set to ViT-Large-224, "Fig.II".The main evaluation metrics adopted for fine-tuning was the Recall score.In Experiment 2, we kept the optimized hyper-parameters frozen and investigated different pre-trained ViT models sizes for image size 224x224 pixels: ViT-base-224, ViT-Base-224-21k, ViT-large-224, ViT-large-224-21k, ViT-Huge-224-21k.

In Experiment 3, with the optimized hyper-parameters frozen and the model set to ViT-Large-224, we investigated the impact of over-sizing by running the same code on two different system architectures. Finally, in Experiment 4 with frozen the hyper-parameters and the model set to ViT-Large-224

TABLE III
RUNPOD SERVER SPECIFICATIONS

System Setup					
System	GPU	VRAM [GB]	RAM [GB]	vCPU	Cost [\$ /hour]
1	1x RTX A4000	16	23	6	0.34
2	1x RTX A5000	24	29	8	0.44
3	1x A40	48	48	9	0.69
4	1x A100 PCIe	80	117	8	1.89

we investigated the impact of fine-tuning in different server providers with similar server architectures, using an NVIDIA A100 GPU, which was the only common GPU card in both setups.

E. Servers Description

ViT Model requires lots of computational resources. In order to have access to them, all the experiments were executed on rented servers from the Runpod provider [38] “Table.III”. In all cases, the PyTorch 2.2 library [39] and the accelerate library [40] were used to run the code. For the server provider comparison, we also used a Google Colab server with a standard account (payed units, no Pro) and an A100 GPU 40 GB.

F. Emission estimation

For emissions and energy consumption calculations, we used the CodeCarbon tool, focusing only on the process of fine-tuning. Given that parts of the process like Python server setup, libraries installations, downloading datasets and models, executed once per server architecture, they were excluded from the energy impact (in order to focus only on the process). Also it was not feasible to get the energy and emissions from parts like setting up the server or installing libraries from the provider. Upon reviewing the CodeCarbon results, we found a discrepancy: the reported total available memory was larger than the maximum allowed by the rented server. This happened because the server is running on containers which make the accurate estimation or used RAM difficult to measure. Since CodeCarbon estimates RAM energy consumption using total available RAM [41] (that is shared to many users), we had to exclude the RAM measurements from the calculation. Consequently, we recalculated emissions manually using CodeCarbon methodology, applying the formula of energy consumed “E” [kWh] multiplied by the carbon intensity “C” [grams of CO₂ per kWh] specific to each country, omitting the RAM portion. The values for carbon intensity are available for each country in the CodeCarbon documentation [42].

G. Environmental metrics

For the environmental impact analysis, four metrics were used. Two are related to the overall impact: 1) Energy consumed without the RAM impact [kWh], 2) Equivalent CO₂ emissions produced [kg CO₂], and two are related to operational efficiency: 3) Power, calculated by dividing the energy (kWh) by the duration (hr) [kW] and 4) Equivalent emission rate [kg CO₂/hour].

TABLE IV
EXPERIMENT 1: ENVIRONMENTAL IMPACT OF HYPER-PARAMETER OPTIMIZATION

Hyperparameter optimization energy consumption (without RAM impact)					
	Energy (kWh)	Duration (hr)	Power (kW)	Emissions (kg CO ₂)	Emission Rate (kg CO ₂ /hr)
E1: 1e-1	0.166	0.747	0.222	0.061	0.082
E1: 1e-3	0.167	0.746	0.224	0.061	0.082
E1: 1e-5	0.167	0.743	0.225	0.061	0.083
E1: 1e-6	0.168	0.75	0.224	0.062	0.082
E2: Constant	0.168	0.748	0.224	0.062	0.082
E2: Cosine	0.168	0.752	0.223	0.062	0.082
E3: b16	0.164	0.728	0.225	0.06	0.083
E3: b32	0.164	0.686	0.239	0.06	0.088
E4: 60 epoc	0.257	0.98	0.263	0.095	0.097
E4: 100 epoc	0.257	0.98	0.273	0.095	0.097
E5: MaxGrad 0.95	0.165	0.741	0.223	0.061	0.082
E5: MaxGrad 0.98	0.166	0.741	0.223	0.061	0.082
E6: WarmUp 100	0.166	0.748	0.222	0.061	0.082
E6: WarmUp 500	0.165	0.748	0.223	0.061	0.081
Total	2.507	10.836	3.232	0.923	1.185
Average	0.179	0.774	0.231	0.066	0.085

H. Limitations

As described above, there are limitations to this case study. 1) We used only the 224x224 image size ViT models. 2) The dataset used is small and consists of 780 samples. 3) The final results include only the energy and emissions from the CPU and GPU used, not RAM. 4) The server location for RunPod was in the USA, while that used for the Google Colab experiment was located in Singapore. This impacts the values of the carbon intensity. 5) For the comparison of pre-trained models, servers with different specifications were used, especially for larger models, to ensure that the models had the necessary resources to be executed. 6) Hyper-parameter optimization was done manually with 14 experiments, rather than running all possible combinations of the 81 hyper-parameter values. 7) Provider comparison was only made on CPU and GPU energy and emission values, as provided by the CodeCarbon tool.

IV. RESULTS

From Experiment 1, we found that the average energy per single hyper-parameter optimization was 0.179 kWh, and the equivalent CO₂ emissions were 0.066 kg. The total energy required for all hyper-parameter optimizations was 2.507 kWh, the total CO₂ emissions 0.923 kg, “Table.IV”.

Almost all hyper-parameter experiment had similar emissions, except for the epoch experiments, where the emissions were 43% higher “Table.IV”. Since duration of each run was different, we divided each value by the duration (in hours) needed to execute in order to investigate whether all hyper-parameter optimizations on the ViT model have the same impact on energy and equivalent emissions. This provides the emission rate of equivalent CO₂ (kg/hr) and the power (kW).

An important decision at this stage was whether to run all possible hyper-parameter combinations using an automated tool or code, or to execute the process manually. We chose to execute the hyper-parameter optimization manually, running each set of scenarios individually, finding the best value for the active hyper-parameter studied in each scenario, and then froze this value for the next set of executions. This approach may not cover all possible combinations of hyper-parameter optimization, which could potentially lead to “Red

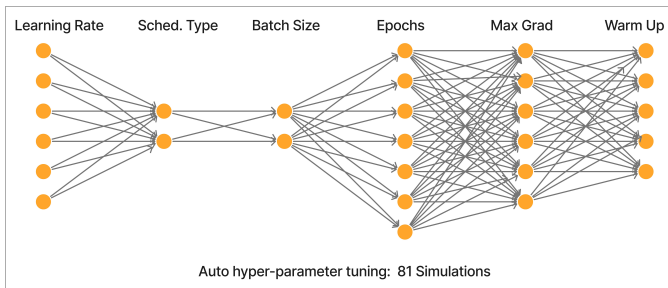


Fig. 3. 81 combinations for hyper-parameter optimization. Learning rate from 1e-1 to 1e-6, Schedule Type: cosine, constant, batch size 16 and 32, Epochs 40 to 100, Max Grad from 1 to 0.95 and warm up from 100 to 500

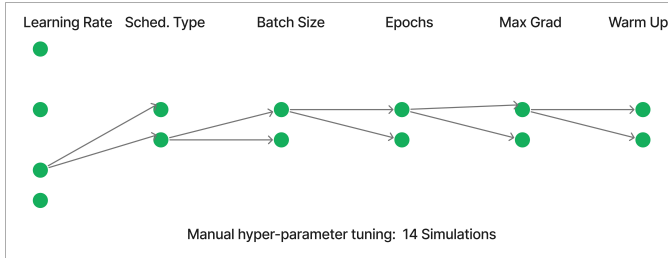


Fig. 4. Manually selecting hyper-parameter for optimization, 14 executions in total. See Table.II for the selected values

Computing” practices, as all combinations would require 81 training executions “Fig.3”. In contrast, our manual approach required only 14 executions “Fig.4”.

In the second set of experiments, we investigated ViT model size on the environmental impact. First, we validated from each separate run that the differences in the recall scores in all model sizes used were not distant, so we could have similar model efficiency. Then, we dived into the emissions and energy study. The ViT-Large-224 and ViT-Large-224-21K models found to perform best on the Benign class, with recall scores of 96.55%. The ViT-Base-224-21K performed best on the Malignant class with recall score of 90.48%, while the ViT-Huge-224-21K achieved the highest recall score of 85.19% for the Normal class “Fig.5”. The explanation of recall scores and model behavior is beyond the scope of this case study, which focuses on environmental impact. However, we found that the dataset was not balanced, with 350 samples in the Benign class, 168 in Malignant, and 106 in Normal, possible leading to inaccurate performance results.

Evaluating the environmental impact, we found that, smaller models produced less equivalent CO₂ emissions during the

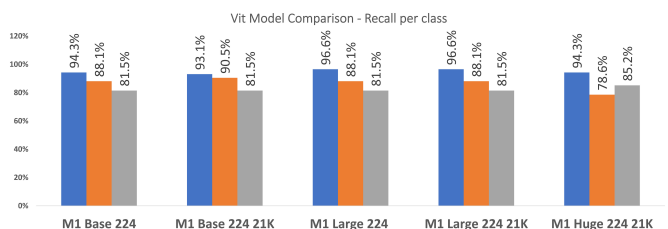


Fig. 5. Google’ ViT pre-trained model comparison. Recall Score per category

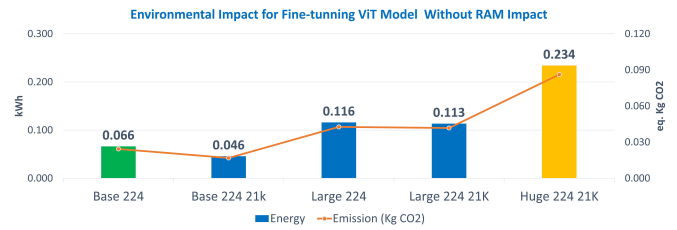


Fig. 6. Energy consumption and the equivalent kg of CO₂ per model. Green: RTX A5000, Blue: RTX A4000, Orange: A40

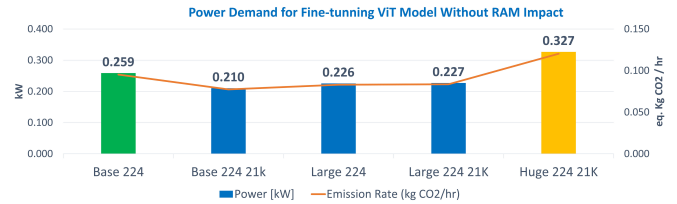


Fig. 7. Power consumption and equivalent rate of kg CO₂ per model. Green: RTX A5000, Blue: RTX A4000, Orange: A40

fine-tuning process. Additionally, the 21K versions showed slightly reduced emissions compared to the base models (without the -21K prefix) “Fig.6”.

Evaluating the efficiency, we observed a different behavior “Fig.7”. The larger model, ViT-Huge-224-21K, exhibited the highest power and emission rate values. The smallest model, ViT-Base-224, followed, with the remaining models showing similar power and emission rate values. It is important to note that the system architecture influences these results, as larger models require more capable servers with larger GPUs for their execution.

The next set of experiments focused on investigating system architecture over-sizing. We executed exactly the same experiment (E5) on an RTX 4000 and a much more capable NVIDIA A100 80GB, both on the same provider (RunPod), “Fig.8”. While the base RTX 4000, capable of executing the code, needed 0.116 kWh, the A100 80GB required 0.126 kWh (8% more). The emissions were 0.043 kg CO₂ and 0.046 kg CO₂ (8% more for the A100), and the execution times were 30.8 minutes for the RTX 4000 and 22.9 minutes for the A100, with the A100 providing faster results (25% improvement). Thus, system over-sizing can play an important role in the resulting emissions during the fine-tuning process of the model. Evaluating the efficiency, the RTX 4000 required 0.226 kW, while the A100 needed 0.338 kW. In terms of emission rates, the RTX 4000 had a rate of 0.083 kg CO₂/hr compared to the A100 with 0.124 kg CO₂/hr. In both efficiency metrics, the

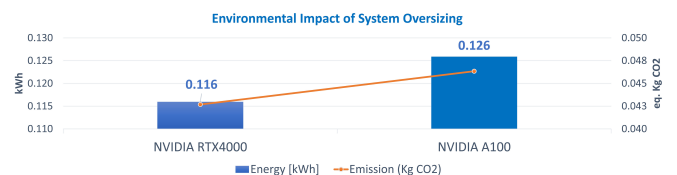


Fig. 8. Impact of system oversize on emissions and energy

TABLE V
NVIDIA GPU CARDS SPECIFICATION USED ON THE TWO PROVIDERS

Nvidia Graphic Cards Specs					
	Graphic Card	Graphic Processor	Cores	TMUS	ROPS
Google Collab	A100 SXM4 40GB	GA100	6912	432	160
Runpod	A100 Pcle 80 GB	GA100	6912	432	160
	Graphic Card	Memory Size	Mem. Type	Bus Width	
Google Collab	A100 SXM4 40GB	40GB	HBM2e	5120 bit	
Runpod	A100 Pcle 80 GB	80 GB	HBM2e	5120 bit	

A100 appeared to have almost 50% higher impact compared to the RTX 4000. Therefore, system over-sizing found to lead to a greater environmental impact and reduced efficiency. Further research on system over-sizing is proposed to provide more accurate results on environmental impact.

Finally, since high-end GPU and TPU systems, required to train and fine-tune ViT models, are too expensive for developers to own, the common practice is to rent such virtual machines or containers through online providers. This leads us to the last research question: whether the choice of online providers impacts the environmental footprint of fine-tuning a ViT model for medical image recognition tasks or not.

To address this, we executed the same code with the optimized hyper-parameters frozen, values of E5 and the model frozen to ViT-Large-224-21K, on both RunPod and Google Colab servers. We chose the NVIDIA A100 because it was the nearest common GPU available between the two providers, with RunPod offering an NVIDIA A100 80GB and Google Colab providing an A100 40GB “Table.V”. The selection of these two providers was based solely on ease of access and the ability to execute the code from our side.

At this point, it is important to note that we have no relationship with either company, and our goal was not to make a direct comparison between the specific two, as many other parameters would be required to perform such task. Also, their systems are not identical. Additionally, it is important to mention that the servers were located in different countries with varying carbon intensity values, as provided by the CodeCarbon tool. We just aimed to see if a significant difference on emissions will occur by running the same code on the two different providers.

Comparing the results of the CodeCarbon tool, we observed that the energy consumption was approximately the same for both providers, ranging from 0.112 kW to 0.126 kW, with a difference of about 10%, ‘Fig.9’. However, the emission rates exhibited a reversed behavior, with an 18% difference in emissions. The carbon intensity values used for the calculations were 488.775 g CO₂/kWh for Singapore (Google Colab) and 368.102 g CO₂/kWh for the USA (RunPod server), year 2022 [42]. That is mostly the main reason of the emission differences. From this comparison we found that the choice of online server provider is mostly important as it is related to the country that servers are located and the energy mixture used by each country.

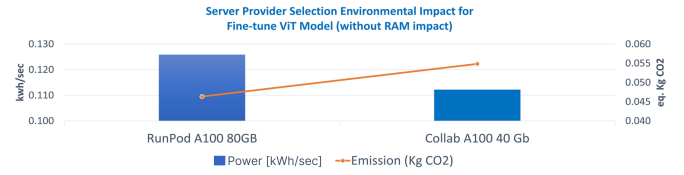


Fig. 9. Impact of system oversize on emissions and energy

V. DISCUSSION

Generally, this case study faced challenges regarding calculating the total maximum available RAM value reported by the CodeCarbon [12] tool. Taking into consideration the complexity of the problem (many parts have to be included as the Graph model of the research [26] presents), it is evident that Green AI needs new and more accurate metrics. Similar findings are also presented in literature. From the results it was found that differences exist on both efficiency and operational metrics studied on this small datasets. Scaling up to large datasets that require much higher resources and much longer execution times can lead to important environmental impacts. The most important findings from our case study are that although the system over-sizing may lead to shorter execution times, it was found to have important environmental impact. Secondly, the server location (and the execution time of day according to literature) selection can reduce the produced emissions. Finally, according to the automation tools that do not apply smart selection of hyper-parameters on each epoch, the manual hyper-parameter estimation can lead to running many simulations that are not needed and lead to bad practices.

Closing we report that the cost for executing all the experiments was 49 €.

VI. CONCLUSIONS

This case study addressed four research questions regarding the environmental impact of fine-tuning Vision Transformer (ViT) models for a specific medical image recognition task. Firstly, it was found that hyper-parameter optimization using the base ViT-Large-224 model consumed an average of 0.179 kWh of energy and produced 0.066 kg of CO₂ emissions, with total optimization requiring 2.507 kWh and 1.185 kg CO₂. Secondly, selecting different pre-trained ViT models revealed that larger models have a higher environmental footprint, while the “-21K” variants slightly reduce this impact compared to their base counterparts. Specifically, the ViT-Huge-224-21K model was the most energy and emission-intensive, followed by the ViT-Base-224, with other models showing similar efficiency. Thirdly, over-sizing the system led to an 8% increase in emissions and a 50% rise in power consumption, although it improved processing speed by 25%. Lastly, the choice of online server provider affected energy consumption by approximately 8% and altered emissions due to server location, highlighting the significant role of server placement in environmental outcomes.

VII. PROPOSALS FOR FUTURE RESEARCH

This case study suggests further investigation in three areas:

- 1) Assessing whether running all possible hyper-parameter combinations is worth the increased environmental impact,
- 2) Conducting experiments on datasets of varying sizes, including the environmental impact of dataset storage,
- 3) Conduct experiments including all code processes from libraries loading up to chart extractions,
- 4) Exploring the environmental impact of online server providers, considering technical aspects, server location, and fine-tuning practices like Flexible Start and Pause and Resume methods that operate during low energy demand periods

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REFERENCES

- [1] Ashish Vaswani, et. al, “Attention Is All You Need,” 31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA, 2017.
- [2] Alexey Dosovitskiy, et. al, “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” ICLR 2021 - 9th International Conference on Learning Representations, 2021.
- [3] Joseph Redmon, et. al, “You Only Look Once: Unified, Real-Time Object Detection”, Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2016.
- [4] David Reinsel, et. al, “The digitization of the world from edge to core,” Framingham: Int. Data Corporation, vol. 16, pp. 1-28, 2018.
- [5] Emma Strubell, Ananya Ganesh, Andrew McCallum, “Energy and policy considerations,” arXiv:1906.02243, 2019.
- [6] Shirin Salehi, Anke Schmeink, “Data-Centric Green Artificial Intelligence: A Survey,” IEEE Transactions on Artificial Intelligence, vol. 5, May 2024.
- [7] You Zhou, et. al, “On the Opportunities of Green Computing: A Survey,” arXiv:2311.00447v3 [cs.AI], 9 Nov 2023.
- [8] Google, “Environmental Report,” 2024. <https://www.gstatic.com/gumdrop/sustainability/google-2024-environmental-report.pdf>.
- [9] Microsoft, “2024 Environmental Sustainability Report,” <https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RW11MJE>.
- [10] Jesse Dodge, et. al, “Measuring the Carbon Intensity of AI in Cloud Instances,” FACCT ’22: Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency, pp. 1877 - 1894, 2022.
- [11] Roy Schwartz, Jesse Dodge, Noah A. Smith, Oren Etzioni, “Green AI,” Communications of the ACM, vol. 63, no. 12, pp. 54-63, 2020.
- [12] “CodeCarbon,” [ulrhttps://codecarbon.io/](https://codecarbon.io/).
- [13] “Carbontracker,” [ulrhttps://carbontracker.org/](https://carbontracker.org/).
- [14] “Green Algorithms,” [ulrhttps://www.green-algorithms.org/](https://www.green-algorithms.org/).
- [15] Lucia Bouza, et. al, “How to estimate carbon footprint when training deep learning models? A guide and review,” Environmental Research Communications, vol. 5, no. 11, 2023.
- [16] Nesrine Bannour, et. al, “Evaluating the carbon footprint of NLP methods: A survey and analysis of existing tools,” Proceedings of the Second Workshop on Simple and Efficient Natural Language Processing, pp. 11–21, 2021.
- [17] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, Li Fei-Fei, “ImageNet: A large-scale hierarchical image database,” IEEE Conference on Computer Vision and Pattern Recognition, pp. 248–255, 2009.
- [18] Eva García-Martín, Crefeda Faviola Rodrigues, Håkan Grahm, “Estimation of energy consumption in machine learning,” Journal of Parallel and Distributed Computing, vol. 134, pp. 75-88, 2019.
- [19] Guha Paul, et. al, “A Comprehensive Review of Green Computing: Past, Present, and Future Research,” IEEE Access, vol. 11, 2023.
- [20] Anindya Bose, Sanjay Nag, “Green Computing – A Survey of the Current Technologies,” Asia-Pacific Journal of Management and Technology, vol. 3, pp. 1-15, 2022.
- [21] Verónica Bolón-Canedo, Laura Morán-Fernández, Brais Cancela, Amparo Alonso-Betanzos, “A review of green artificial intelligence: Towards a more sustainable future,” Neurocomputing, vol. 599, p. 128096, 2024.
- [22] Tanja Tornede, Alexander Tornede, Jonas Hanselle, Marcel Wever, Felix Mohr, Eyke Hüllermeier, “Towards Green Automated Machine Learning: Status Quo and Future Directions,” Journal of Artificial Intelligence Research, vol. 77, pp. 427-457, 2023.
- [23] Adrien Berthelot, Eddy Caron, Mathilde Jay, Laurent Lefèvre, “Estimating the environmental impact of Generative-AI services using an LCA-based methodology,” 31st CIRP Conference on Life Cycle Engineering (LCE 2024), pp. 707–712, 2024.
- [24] Christian Clemm, Lutz Stobbe, Kishan Wimalawarne, Jan Druschke, “Towards Green AI: Current Status and Future Research,” presented at the Electronic Goes Green 2024+ Conference.
- [25] Yehia Ibrahim Alzoubi, Alok Mishra, “Green artificial intelligence initiatives: Potentials and challenges,” Journal of Cleaner Production, vol. 468, p. 143090, 2024.
- [26] Michael Färber, David Lamprecht, “The Green AI Ontology: An Ontology for Modeling the Energy Consumption of AI Models,” 21st International Semantic Web Conference (ISWC 2022), Virtual Event, pp. 1-8, October 23–27, 2022.
- [27] Stefanos Georgiou, Maria Kechagia, Tushar Sharma, Federica Sarro, Ying Zou, “Green AI: Do Deep Learning Frameworks Have Different Costs?,” in IEEE/ACM 44th International Conference on Software Engineering (ICSE), 2022.
- [28] Negar Alizadeh, Fernando Castor, “Green AI: A Preliminary Empirical Study on Energy Consumption in DL Models Across Different Runtime Infrastructures,” in CAIN 2024: Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering - Software Engineering for AI, 2024.
- [29] David Patterson, Joseph Gonzalez, Quoc Le, Chen Liang, Lluís-Miquel Munguia, Daniel Rothchild, David So, Maud Texier, Jeff Dean, “Carbon Emissions and Large Neural Network Training,” Environmental Science, Computer Science, Engineering, 2021.
- [30] Kai Huang, Hanyun Yin, Heng Huang, Wei Gao, “TOWARDS GREEN AI IN FINE-TUNING LARGE LANGUAGE MODELS VIA ADAPTIVE BACKPROPAGATION,” International Conference on Learning Representations (ICLR 2024).
- [31] Dagoberto Castellanos-Nieves, Luis García-Forte, “Strategies of Automated Machine Learning for Energy Sustainability in Green Artificial Intelligence,” MDPI Applied Sciences, 2024.
- [32] Andrea Asperti, Davide Evangelista, Elena Loli Piccolomini, “A Survey on Variational Autoencoders from a Green AI Perspective,” SN Computer Science, vol. 2, no. 301, 2021.
- [33] David Patterson, et. al, “The Carbon Footprint of Machine Learning Training Will Plateau, Then Shrink,” IEEE Computer, vol. 55, 2022.
- [34] emre570/breastcancer-ultrasound-images, [Online]. Available: <https://huggingface.co/datasets/emre570/breastcancer-ultrasound-images>.
- [35] H. Face, “Transformers,” [Online]. Available: <https://huggingface.co/docs/transformers/index>.
- [36] N. Raw, “Hugging Face Tutorial on Fine Tune Vit Model,” 2024. <https://huggingface.co/blog/fine-tune-vit>.
- [37] Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in Proceedings of NAACL-HLT, Minneapolis, 2019.
- [38] “RunPod,” <https://RunPod.com>.
- [39] “Pytorch Library,” <https://pytorch.org/>.
- [40] “Hugging Face Accelerate Lib,” [Online]. Available: <https://huggingface.co/docs/accelerate/index>.
- [41] “CodeCarbon RAM Methodology,” https://mlco2.github.io/codecarbon/search.html?q=ram&check_keywords=yes&area=default.
- [42] “CodeCarbon Country Intensities,” [Online]. Available: https://github.com/mlco2/codecarbon/blob/master/codecarbon/data/private_infra/global_energy_mix.json.
- [43] “techpowerup.com,”: <https://www.techpowerup.com/gpu-specs>
- [44] Shashank Nag, et. al, “Lightweight Vision Transformers for Low Energy Edge Inference”, ML for Computer Architecture and Systems (MLArchSys), ISCA 2024.